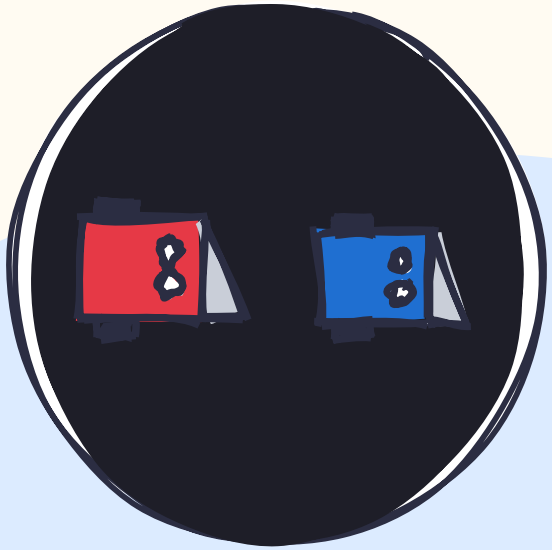
Young Innovators · Class 25

Robot Sumo Challenge

Find. Push. Survive. Autonomous!

Physics | Sensors | State machine



Did you know?
In Japan, robot sumo has run since 1989, and the fastest mini-sumo robots can win a match in under half a second.

Let's think...

Small robot vs big robot: who wins?

David vs Goliath

Two robots in a ring, nobody holding a remote. The rule is simple: push the other one out. Is the biggest motor always the winner? Or can a clever, low, wedge-shaped robot win?



Biggest motor!

Heaviest!

Smartest design!

Today: you build and program an **autonomous sumo robot** and compete in a class tournament.

Learning objectives

By the end of this class I can...

1	**Pushing physics:** explain how **weight, grip, a wedge** and a low **centre of gravity** win pushing matches	**Proof:** I can say why my robot's design helps it push
2	**Sense the ring:** use **IR line sensors** to detect the white edge and **ultrasonic** to find the opponent	**Proof:** my robot never drives out by itself
3	**State machine:** program **SEARCH → ATTACK → ESCAPE** states and compete in a tournament	**Proof:** my robot finds, pushes and survives a full round

Objective 3 · State machine · video

Video time!



The Freakishly Fast World of Robot Sumo

Great Big Story · about 3 minutes · youtu.be/mS-L2fpV1Is

Watch for...

1. How fast does a robot sumo match **end**?
2. What **shape** do the winning robots have?
3. Why are some robots so **low and flat**?

Scan to watch

After the video: What is one design idea from the video you will copy for your robot?

Objective 1 · Pushing physics

The physics of pushing

The idea

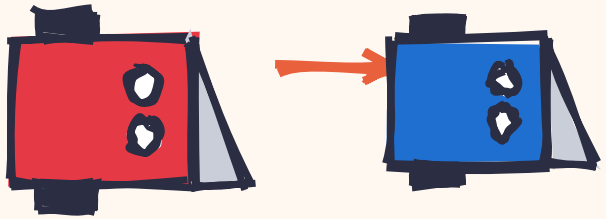
To push, your wheels need **grip**. Maximum push ≈ **weight × grip** (friction). Heavier robot + grippy tyres = more push. Fast spinning wheels that slip push nothing!

A **wedge** scoops under the opponent, lifting their wheels so **they** lose grip.

A **low centre of gravity** stops you being tipped over.

Class rules: max **500 g**, 15 × 15 cm.

Example: a tug of war



wedge + weight + grip

In tug of war, the team that digs their feet in and leans low wins, not the team that pulls fastest. Grip and stance beat raw speed.

Objective 2 · Sense the ring

Two kinds of eyes

The idea

IR line sensors (TCRT5000) underneath, at the front: they shine infrared down. The black ring absorbs it; the **white edge** reflects it back. White seen = **danger, about to fall out!**

Ultrasonic (HC-SR04) at the front: measures distance. Object within ~**40 cm** (the ring is 77 cm) = **opponent ahead!**

Edge beats everything: never chase an opponent off the ring.

Example: walking near a pool



white = 1

black = 0

You watch your feet near the pool edge and look up to find your friend. Survival first (don't fall in), then the game.

Objective 3 · State machine

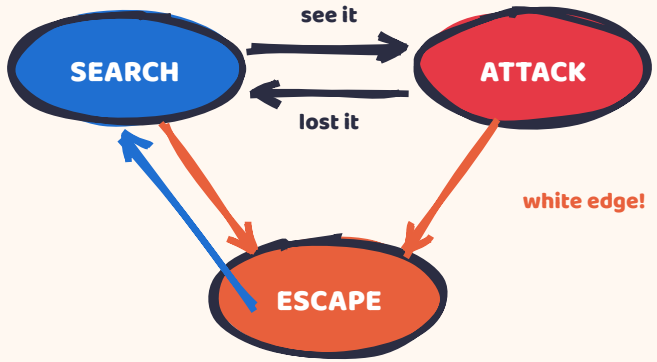
Think in states

The idea

Instead of one giant if-tangle, the robot is always in one **state**:
SEARCH: spin slowly, looking with ultrasonic.
ATTACK: opponent seen → full speed forward.
ESCAPE: white edge seen → reverse, then turn away.

Each loop: check sensors → maybe change state → act. ESCAPE has the **highest priority**.

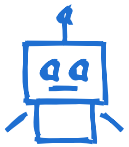
Example: a goalkeeper



A goalkeeper is either watching (search), diving (attack) or getting back into the goal (escape). They switch state based on what they see.

Real world

Autonomous robots that decide



Robot vacuums

Bump → reverse → turn: an ESCAPE state!



Self-driving cars

Cruise, follow, brake, overtake states



Game enemies

Patrol, chase, flee: same state machines



Factory robots

Stop instantly when a human enters (highest priority)

? Which state in your daily routine has the 'highest priority'?

Japan's All-Japan Robot Sumo

Thousands of teams compete every year, from school students to professional engineers.

Quick quiz

Sumo science

Question	Answer
1. Two ways to get more push?	
2. Why does a wedge help?	
3. IR sensor reads white. Which state?	
4. Ultrasonic reads 25 cm. Which state?	
5. Both happen at once. Which wins?	

Answers

Cover! 1: more weight, grippier tyres · 2: lifts the opponent so they lose grip · 3: ESCAPE · 4: ATTACK · 5: ESCAPE

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human state machine
2. Our sumo kit (per team)
3. Sensor and motor helpers
4. The state machine
5. Tune, then tournament
6. Match log
7. Sumo strategist challenges

Class game

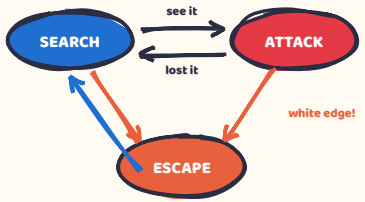
Human state machine

How to play

1. Pairs. One is the **robot** (eyes closed, arms out), one is the **sensor caller**.
2. Caller shows a card: **NOTHING**, **TARGET** or **EDGE**.
3. Robot acts: NOTHING = turn slowly on the spot, TARGET = step forward, EDGE = step back and turn.
4. Speed up! Then the caller shows TARGET + EDGE together. What should happen?

Then discuss

Why must EDGE always win?



Big idea

One state at a time, highest priority first.

Build it

Our sumo kit (per team)



2WD robot chassis
From earlier projects



Arduino Uno + L298N
Brains + motor driver



2 × TCRT5000 IR
Front left + front right, underneath



HC-SR04
Front, facing forward



Card / plastic wedge
Plus coins or washers for weight



Ring
77 cm black circle, 5 cm white edge

Pins: IR L → D2, IR R → D3. Trig → 10, Echo → 11. L298N ENA 5, IN1 6, IN2 7, ENB 9, IN3 8, IN4 4.

Fair play
Robots must be autonomous: no remotes. No spikes, sharp edges, glue or anything that damages the ring or other robots.

MAKE

Objective 2 · Sense the ring · Build step 1 of 3

Sensor and motor helpers

```
const int IRL = 2, IRR = 3, TRIG = 10, ECHO = 11;
const int ENA = 5, IN1 = 6, IN2 = 7, ENB = 9, IN3 = 8, IN4 = 4;

void drive(int left, int right) {          // -255 ... 255
  digitalWrite(IN1, left > 0);  digitalWrite(IN2, left < 0);
  digitalWrite(IN3, right > 0); digitalWrite(IN4, right < 0);
  analogWrite(ENA, abs(left));  analogWrite(ENB, abs(right));
}

bool edge() {                              // white = HIGH on most modules
  return digitalRead(IRL) == HIGH || digitalRead(IRR) == HIGH;
}

int cm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long t = pulseIn(ECHO, HIGH, 20000);
  return t == 0 ? 999 : t * 0.0343 / 2;
}
```

drive()

One function for both motors: drive(200, 200) forward, drive(-150, 150) spin.

Test edge

Hold the robot over white, then black. Print edge(). Swap HIGH/LOW if your module is opposite.

999

No echo = nothing in range, so we return a big number.

MAKE

Objective 3 · State machine · Build step 2 of 3

The state machine

```
enum State { SEARCH, ATTACK, ESCAPE };
State state = SEARCH;

void setup() {
  pinMode(IRL, INPUT); pinMode(IRR, INPUT);
  pinMode(TRIG, OUTPUT); pinMode(ECHO, INPUT);
  int motorPins[] = {ENA, IN1, IN2, ENB, IN3, IN4};
  for (int p : motorPins) pinMode(p, OUTPUT);
  delay(5000);                          // RULE: 5-second start delay
}

void loop() {
  // 1. choose the state (highest priority first)
  if (edge())             state = ESCAPE;
  else if (cm() < 40)     state = ATTACK;
  else                    state = SEARCH;

  // 2. act
  switch (state) {
    case ESCAPE: drive(-220, -220); delay(300);   // back off
                 drive(200, -200);  delay(250);   // turn away
                 break;
    case ATTACK: drive(255, 255); break;          // full push
    case SEARCH: drive(140, -140); break;         // spin and look
  }
}
```

Priority

edge() is checked FIRST, so escaping always wins.

5 s delay

Sumo rule: both robots wait 5 s after the start signal.

Tune

Spin speed too fast? The ultrasonic can't keep up. Try 100 to 160.

Objective 3 · Build step 3 of 3

Tune, then tournament

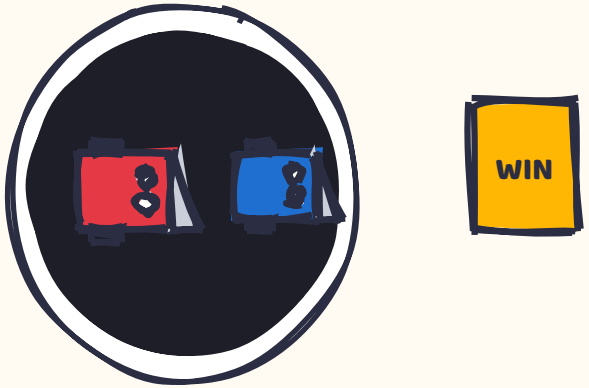
a **Edge test**: place the robot alone in the ring for 30 s. It must never leave.

b **Target test**: put a box in the ring. Does it find it and push it out?

c Add the **wedge** and **weight** (stay under 500 g). Lower = better.

d Tournament: best of 3 rounds, 60 s each. Place robots back-to-back, start together.

e After each match: what would you change? One fix per round!



Check: Your robot stays in alone and pushes a box out.

Keep hands out of the ring while robots are live. Pick up a robot only after it is switched off.

Test like an engineer

Match log

Test	Result	Notes
Alone 30 s: stayed in?		
Pushed the box out (time)		s
Round 1 vs		W / L
Round 2 vs		W / L
Best fix we made		

Debug checklist

Drives out of the ring? IR logic reversed, or sensors too high (2 to 8 mm above the floor).
Spins forever? Ultrasonic too high or angled: aim it at robot height.
Resets when motors start? Use a separate battery for motors.

Engineers improve one thing at a time, then test again. Change two things and you won't know which helped.

MAKE

Level up!

Sumo strategist challenges

Choose your challenge



Two eyes: add a second ultrasonic to know if the opponent is left or right.



Side attack: circle around first, then attack from the side.



Defence: if pushed backwards, turn sideways to escape the push.



Smart search: spin left first if you last saw them on the left.

Think like an engineer

Why might a slower robot with more grip beat a faster one?

Boss level

Make the robot detect that it is being pushed (wheels forward but not moving) and react.

Objective check

Can I really do it? Prove it.

01 Explain three design choices that help your robot push.

02 Show your robot alone in the ring for 30 s: it stays in. How?

03 Point to SEARCH, ATTACK and ESCAPE in your code. Why is ESCAPE first?

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Tournament: everyone competes. Awards for Champion, Best Design and Best Comeback Fix!

Show what you know

What did we learn?

- [x] Push ≈ weight × grip; wedges steal grip
- [x] A low centre of gravity resists tipping
- [x] IR sees the white edge; ultrasonic finds the opponent
- [x] State machines: SEARCH, ATTACK, ESCAPE
- [x] Highest priority first: survival beats attack

Exit ticket: your IR sensor sees white AND the opponent is 20 cm away. What does your robot do?

At home

Watch a robot vacuum or a game enemy. Can you spot its states?



Next class: Walking Linkage Robot!